Knowing through Scientific Models

Course B-KUL-W0Q07A
KU Leuven, Winter Semester 2026/27
Mondays, 9:00-12:00
MSI 1 - 00.20

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1. Course Description

Much of epistemology focuses on learning through perception, testimony, and different models for rational belief. But the world is complicated and actual production of knowledge rarely operates at these either very simple or highly formalized levels. Scientists, but also business executives, politicians, or sports enthusiasts use different kinds of *models* and *representations*. But how do these substitutions, with their simplifications, mutual contradictions, or outright falsehoods, enable us to understand and know about the world? Understanding the functioning and limitations of modelling becomes more pressing through the ongoing digitization of more and more branches of social and economic life.

This course aims to map the state of current debates of thinking about scientific representations and mathematical models in particular. Knowledge of general philosophy of science is welcome but not required. Post-secondary scientific education and knowledge of the philosophy of special sciences are not required.

2. Mode of Instruction

The course combines a weekly lecture with seminar discussion. Students submit **reading reflections (see below)**. The syllabus and reading list are presented during the first lecture. All required readings and – where possible – the recommended readings, are shared through Toledo.

3. Technology in the Classroom

While you should be able to use whatever note taking system works for you, your peers also have a right to not be distracted. Illuminated screens compete for attention and distract not only the user, but everyone in view. Wherever possible, I am therefore asking you to use paper or eInk devices to bring and take notes. All slides will also be made available online, so you will not have to write very much. I am also asking you to leave smartphones out of view and reserve the right to ask you to leave the classroom if you need to use a phone. Students with accommodations are of course excepted, and you can also ask me directly if you would like to make an exception to this policy.

4. Assessment

Your evaluation comprises weekly submissions, an essay assignment, and an oral examination.

4.1 Weekly Submissions (20%)

Throughout the course, you are expected to submit a reading reflection of **100-200 words before 12:00 (noon) on Fridays.** They must be submitted to gregor.boes@kuleuven.be. The email title must be exactly "Reading Reflection Week XX Surname, Given Name", for example "**Reading Reflection Week 02 Doe, Jane**". Please send **only text in the email body**, no attachments.

Here you show that you engaged with the assigned reading and formulate questions or challenges. You can refer to other literature or your own research interests, but it should be clear that

you have engaged with the assigned reading. These reflections will be the basis for discussion in class, and you may be asked to read your reflection or a of it in class. If you do not want your name to appear next to your reading reflection, please state so explicitly after before or after each submission.

A complete assignment receives 2 points, an insufficient attempt or late submission receives 1 point, non-submission a 0. The maximum number of points is capped at 20. No use of generative writing technologies or copying from published sources is allowed. Since this is part of your examination, suspected breach of this requirement will be reviewed as potential academic misconduct and may disqualify you for completing the course. Submitting fewer than 50% of all reading reflections disqualify you from completing the course.

4.2 Final Exam with Essay (80%)

Since this is a masters-level course, the final exam is focused on high-level, synthesizing and original tasks, instead of the reproduction of studied answers.

Over the course of four hours, students are submitting written responses to a range of questions that cover the entirety of the course. Answers will consist of free texts of 100-300 words per question.

A final section, worth 40% of the exam score, will be based on a **free-form essay of 1,000 +/- 100** words on a topic that the student can choose freely and prepare in advance.

The essay should be a succinct, self-contained piece of writing that explores topics from one or two sessions, rather than the entire course. Your goals should be to write an essay that addresses a well-specified and relevant question, has a logical structure, is clearly written, and both charitable and critical in its engagement with primary and secondary literature.

The essay topic should be decided and researched *beforehand*. Students can bring a printout of 1 double-sided A4 paper, Times New Roman 10pt, 2.5cm margins, with quotations and literature references to the examination. This document cannot contain prewritten portions of the essay and has to be submitted at the end of the exam. There will be a 1h mock session in which students can practice writing under such conditions.

All submissions are graded **anonymously**, please ensure that your name is not stated anywhere in your submission.

4.3 Grading Rubric

Reading reflections are graded as good (2 points) or acceptable (1 point), where late submissions will only achieve an 'acceptable' score.

Exam responses will be graded according to the general grading rubric of KU Leuven:

18-20	**A+**	**Excellent**	high distinction, work of excellent quality
16-17	**A**	**Outstanding**	work of very good quality and well above average
15	**A-**	**Very Good**	distinction, work of good quality
14	**B+**	**Good**	generally competent work of quality
13	**B**	**Satisfactory**	fair but with some shortcomings
12	**C+**	**Sufficient**	fair but with shortcomings
11	**C**	**Barely Sufficient**	performance meets the minimal criteria
10	**D**	**Barely Sufficient**	performance just meets the minimal criteria
8-9	**F**	**Insufficient**	some more work is required to obtain a passing grade
6-7	**F**	**Very Insufficient**	more work is required to obtain a passing grade
0-5	**F**	**Totally Insufficient**	considerable further work is required to obtain a passing grade

For the essay, notice that it is an examination document, not a literary piece of writing. For substantial pieces of writing of 1,000 words or more, the grade is composed of the following factors:

Factor	Weight	Description
Structure	20%	Different parts of the essay are clearly related and no parts are redundant. The introduction sets up the topic efficiently. The flow of

		argumentation is presented transparently. It is clear throughout what the essay aims to do at the current point.
Clarity	30%	The essay is self-contained. Thoughts are expressed precisely and not obscured by language. Relevant distinctions are drawn, specialized terms are introduced and ambiguities dissolved and anticipated.
Scholarship	30%	Sources are referenced correctly and consistently (in whichever bibliographic style you prefer, as long as you add a separate list of references). Reconstruction of authors is efficient, fair, and critical. Where necessary, limitations and presuppositions of the presented argument are spelled out.
Novelty	20%	The essay does not need to make a publishable research contribution, but it clearly exceeds a literature review and demonstrates the student's own thought and engagement with the material discussed.

Points may be deducted for exceeding the permitted word count. Shorter pieces of writing are assessed without explicit grades for individual factors, but with the same emphasis on structure, clarity, and demonstration of scholarship.

5. Workload

When preparing your weekly schedule, consider the additional time you will need to complete assignments and prepare readings for this class. Assigned readings often need more than one pass, and you may want to consult the recommended readings to deepen your understanding.

6 ECTS correspond to an expected workload of 150-180 hours, including

12 x 3h attendance
11 x 4h (est.) review and preparatory reading
11x 1h (est.) writing for continuous assessments
30h (est.) revision for final exam
30h (est.) research for written assignment
5h exam and mock exam attendance

for a total of 156h.

6. Midterm Feedback

Towards the middle of the course, you will receive a webform to evaluate the course up to this point. Where possible, suggestions are taken into account for the rest of the course.

7. Weekly schedule

There are two kinds of readings: **required** (in bold) and recommended. Everyone is expected to study required readings **before** coming to class. Recommended readings are optional but will deepen your understanding and may be discussed in the lecture.

English versions of required readings are made available through Toledo. Where applicable, feel free to consult them in the original language.

Wk	Date	Topic	Readings (**required** | recommended)
1	21 Sep 26	Introduction	**This Syllabus**
2	28 Sep 26	Structuralism	**van Fraassen 2008, *Scientific Representation*, ch. 8** French 2020, *There Are No Such Things As Theories*, ch. 2 Muller 2011, 'Reflections on the Revolution at Stanford' Friedman 2011, *A Parting of the Ways*, ch. 7 Cassirer 1910, *Substanzbegriff und Funktionsbegriff. Untersuchungen über die Grundfragen der Erkenntniskritik*
3	5 Oct 26	Idealization	**Elliott-Graves and Weisberg 2014, 'Idealization'** McMullin 1985, 'Galilean Idealization' Strevens 2016, 'How Idealizations Provide Understanding' Potochnik 2017, *Idealization and the Aims of Science* Elgin 2017, *True Enough* Spagnesi 2023, 'Regulative Idealization'
4	12 Oct 26	Mathematization	**Nguyen and Frigg 2021, 'Mathematics Is Not the Only Language in the Book of Nature'** Bueno and Colyvan 2011, 'An Inferential Conception of the Application of Mathematics' Pincock 2012, *Mathematics and Scientific Representation* Vos 2022, 'Structuralism and the Quest for Lost Reality' Wigner 1960, 'The Unreasonable Effectiveness of Mathematics in the Natural Sciences' Husserl 1970, *The Crisis of European Sciences and Transcendental Phenomenology*, esp. §9 Contessa 2010, 'Empiricist Structuralism, Metaphysical Realism, and the Bridging Problem'
5	19 Oct 26	Models	**Morgan and Morrison 1999, *Models as Mediators*, ch.2: Models as Mediating Instruments**

			Weisberg 2013, *Simulation and Similarity* Ch. 2: Three kinds of Models Teller 2001, 'Twilight of the Perfect Model Model' Parker 2020, 'Model Evaluation' Bokulich and Parker 2021, 'Data Models, Representation and Adequacy-for-Purpose'
6	26 Oct 26	Gricean deflation	**Callender and Cohen 2006, 'There Is No Special Problem About Scientific Representation'** Ruyant 2021, 'True Griceanism' Grice 1991, *Studies in the Way of Words*, chs. 6 & 14 Husserl 2001, *Logical Investigations*, first investigation
7	2 Nov 26	Inferentialism	Suárez 2004, 'An Inferential Conception of Scientific Representation' Giere 2010, 'An Agent-Based Conception of Models and Scientific Representation' Suárez 2003, 'Scientific Representation' Suárez 2024, *Inference and Representation*
8	9 Nov 26	No Session	**No Session after thesis week**
9	16 Nov 26	Artistic Representation	**Elgin 2010, 'Telling Instances'** Goodman 1976, *Languages of Art*, secs. I.1-2, II.1-4 Merleau-Ponty 1964, 'Eye and Mind' Perini 2005, 'The Truth in Pictures' Bueno and French 2011, 'How Theories Represent' Sartori 2025, 'Why We Love Pictures (for the Wrong Reasons)'
10	23 Nov 26	Metaphor and Fiction	**Frigg and Nguyen 2016, 'The Fiction View of Models Reloaded'** Toon 2012, *Models as Make-Believe*, ch. 3 Hesse 1963, *Models and Analogies in Science*, ch. 2

			Cartwright 2010, *The Dappled World*, ch. 2 Rouse 2015, *Articulating the World*, ch. 9
11	30 Nov 26	Artefactualism	**Knuuttila 2011, 'Modelling and Representing: An Artefactual Approach to Model-Based Representation'** Frigg and Nguyen 2020, *Modelling Nature*, ch. 8, ch. 9 Rheinberger 2005, 'Gaston Bachelard and the Notion of "Phenomenotechnique"'
12	7 Dec 26	History and Philosophy of Measurement	**Tal 2024, 'Models and Measurement'** Basso and Lisciandra 2024, 'Models and Measurement of Inequality' Gelfert 2024, 'Practice-Oriented Approaches to Scientific Modeling' Ruthenberg and Chang 2017, 'Acidity' Chang 2004, *Inventing Temperature* Krantz 2007, *Foundations of Measurement. 1* Tal 2017, 'Calibration'
13	14 Dec 26	Machine Learning Models	**Sullivan 2024, 'Do Machine Learning Models Represent Their Targets?'** Ruyant 2026, 'What Do Large Language Models Represent?' Freiesleben et al. 2024, 'Scientific Inference with Interpretable Machine Learning' Ratti et al. 2026, 'The Modelling Structure of AI-Based Science', ch.3 Buijsman and Durán 2024, 'Epistemic Implications of Machine Learning Models in Science' Räz and Beisbart 2024, 'The Importance of Understanding Deep Learning'
	15 Jan 27, TBD	**FINAL EXAM in VHI – 02.22 (klas D2) VHI – 02.22 (klas E2)**	**4h in-person exam.** **Comprehension questions about the whole breadth of the course,** **plus opportunity to pick a specialized topic for a short essay.**

References

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